

ABHISEK NIRAULA

7067142405

108 College Station Road, Athens, GA, USA

nabhisek02@gmail.com

RESEARCH INTERESTS: Poultry Diseases, Immunology, Gut Health, Parasitology, Microbiome Bioinformatics

EDUCATION

University of Georgia

Athens, GA, USA

PhD in Poultry Science (Immunology and Health)

Jan. 2025-Present

Masters in Poultry Science (Immunology and Health)

Aug. 2022-Dec. 2024

Thesis: Differential response of divergent genetic lines of chickens to mixed *Eimeria* infection

Agriculture and Forestry University, Nepal

Dec. 2014-Oct. 2020

Bachelor of Veterinary Science and Animal Husbandry (equivalent to DVM)

Thesis: Knowledge, attitude, and practices (KAP) pertaining to zoonotic diseases among smallholder livestock owners of the Tulsipur region in Nepal

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug. 2022-Present

Avian Immunobiology Lab, University of Georgia

Athens, GA, USA

- Planned and conducted in vivo and in vitro experiments
- Conducted microbiome data analysis using the QIIME2 platform
- Trained in harvesting intestinal organoids from day-old chicks/poults
- Conducted RNA/DNA extraction, qPCR, ELISA, and lesion scoring (gross/microscopic)
- Analyzed data using JMP Pro, Python, and R
- Mentored undergraduate students in lab techniques, data documentation, and analysis

RESEARCH FUNDING

Dalloul, R. A., **Niraula, A.**, Fuller, A. L. "Cecal organoids offer a reliable laboratory model to evaluate alternative chemotherapeutic compounds against *Histomonas meleagridis*." US Poultry & Egg Association, 2026-2029.

Niraula, A. "Gastro-intestinal parasite burden in Golden Jackal population and its implication in public health". Zebra Foundation for Veterinary Zoological Education, 2020.

PUBLICATIONS

1. **Niraula, A.**, Fuller, A. L., & Dalloul, R. A. (2026). **Invited Review.** Histomoniasis in poultry: Transmission biology, experimental models, and alternative strategies for prevention and control. In preparation. To be submitted to Avian Dis.
2. **Niraula, A.**, Blue, CEC., Calik, A., Ribeiro, G. C., Dastjerdi, P. Z., & Dalloul, R.A. (2026). Phytogenic blends positively modulate ileal and cecal microbiota during necrotic enteritis challenge in chickens. In preparation.

3. **Niraula, A.**, Calik, A., Fenster, D.A., Wagner, A.L., Médina, B., Girard, I., Fuller, A.L. & Dalloul, R.A. (2026). Protective effects of phytogenic supplementation in turkey poult raised in a floor pen lateral transmission model of *Histomonas meleagridis*. To be submitted to Poultry Science.
4. Subedi, D., Mallavarapu, B., Patil, P., **Niraula, A.**, Kandel, M., Kumar, S., and Singh, A.K. (2026). Sources, transmission, and control of *Salmonella* in vertically integrated broiler chicken production. Submitted to Comprehensive Reviews in Food Science and Food Safety.
5. Subedi, D., Sameer T., Madhav P., Gurung, P., Kafle, S., Bhattarai, S., **Niraula, A.**, Marasini, H., Kandel, M., Karki, S. & Tiwari, A. (2026). A global review of highly pathogenic avian influenza (HPAI) and control strategies in Nepal. Zoonotic Diseases 6(2), 11.
6. Fenster, D. A., Qudsieh, R., Wang, J., Calik, A., **Niraula, A.**, & Dalloul, R. A. (2026). Effects of a direct fed microbial and xylanase enzyme blend on *Salmonella* Typhimurium colonization, antioxidant status, and performance in broiler chickens. Poult. Sci., 105, 106684.
7. **Niraula, A.** and Dalloul, R. A. (2025). Genetic basis of disease resistance to coccidiosis and necrotic enteritis. Poult. Sci. 105, 106169.
8. **Niraula, A.**, Wolc, A., Fulton, J. E., Taylor Jr, R. L., & Dalloul, R. A. (2025). The chicken major histocompatibility complex (MHC-B) and alloantigen systems A, D, E, and I impact resistance to coccidiosis. Poult. Sci. 104, 105151.
9. Calik, A., **Niraula, A.**, Dong, B., Blue, C.E.C., Fenster, D.A. & Dalloul, R. A. (2024). Iohexol-based assessment of intestinal permeability in broilers challenged with *Eimeria maxima*, *Clostridium perfringens* or both. Front. Physiol. 15:1520346.
10. Subedi, D., Farhan, M.H.R., **Niraula, A.**, Shrestha, P., Chandran, D., Acharya, K.P. & Ahmad, M. (2024). Avian influenza in low and middle-income countries (LMICs): outbreaks, vaccination challenges, and economic impact. Pak. Vet. J., 44(1): 9-17.
11. Subedi, D., Chandran, D., **Niraula, A.**, Paudel, S., Acharya, K. P., & Dhama, K. (2022). Global Climate Change, Floods, and Associated Zoonotic Disease Outbreaks: A mini-review insight. Indian Vet. J., 99(11): 26-32.
12. **Niraula, A.**, Sharma, A., & Dahal, U. (2021). Knowledge, attitude, and practices (kap) regarding zoonotic diseases among smallholder livestock owners of Tulsipur sub-metropolitan city, Nepal. J. Healthc. Dev. Ctries., 1: 41-46.

ORAL AND POSTER PRESENTATIONS

1. **Niraula, A.**, Fenster, D.A., Ribeiro, G. C., Dastjerdi, P. Z., Fuller, A.L., Dalloul, R.A. (2026, Jul. 10-13). Cecal microbiome profiles in directly inoculated and laterally exposed turkey poult during *Histomonas meleagridis* challenge [Oral Presentation-Abstract accepted]. Poultry Science Association Annual Meeting, Toronto, Canada.
2. Ribeiro, G. C., Fenster, D.A., Dong, B., **Niraula, A.**, Dastjerdi, P. Z., Tavares, M. N., Dalloul, R.A. (2026, Jan. 26-27). Yeast-derived additives improve intestinal morphology in broilers under a necrotic enteritis challenge [Oral Presentation-Abstract accepted]. Poultry Science Association Annual Meeting, Toronto, Canada.
3. **Niraula, A.**, Fenster, D.A., Rhodes, R., Dastjerdi, P. Z., Ribeiro, G. C., Dalloul, R.A. (2026, Jan. 26-27). Coccidiosis vaccination on the day of hatch modulates gut integrity markers [Poster Presentation]. International Poultry Scientific Forum, Atlanta, GA, USA.
4. Rhodes, R., **Niraula, A.**, Fenster, D.A., Dastjerdi, P. Z., Ribeiro, G. C., Dalloul, R.A. (2026, Jan. 26-27). Impact of day of hatch coccidiosis vaccination on performance parameters and immune markers [Poster Presentation]. International Poultry Scientific Forum, Atlanta, GA, USA.
5. Ribeiro, G. C., Fenster, D.A., Dong, B., **Niraula, A.**, Dastjerdi, P. Z., Rodriguez, M. A., Dalloul, R.A. (2026, Jan. 26-27). Effects of yeast-derived dietary additives on performance and immune

- responses of broilers during a subclinical necrotic enteritis challenge [Oral Presentation]. International Poultry Scientific Forum, Atlanta, GA, USA.
6. **Niraula, A.**, Blue, CEC., Suarez, M.G., Nacer-Khodja, E., Dalloul, R.A. (2025, Jul. 29-31). Sulfated polysaccharides positively influence the cecal microbiota of broilers during necrotic enteritis challenge [Poster Presentation]. American Association of Avian Pathologists Annual Meeting, Portland, OR, USA.
 7. Dalloul, R.A., and **Niraula, A.** (2025, Jul. 14-17). Host genetics associated with coccidiosis and necrotic enteritis [Symposium Presentation]. Poultry Science Association Annual Meeting, Raleigh, NC, USA.
 8. **Niraula, A.**, Blue, CEC., Wagner, A.L., Médina, B., Girard, I., Dalloul, R.A. (2025, Jul. 14-17). Phytogenic supplementation modulates the ileal and cecal microbiota supporting gut health during a necrotic enteritis challenge [Oral Presentation]. Poultry Science Association Annual Meeting, Raleigh, NC, USA.
 9. **Niraula, A.**, Fenster, D.A., Blue, CEC., Wagner, A.L., Médina, B., Girard, I., Fuller, A.L., Dalloul, R.A. (2025, Jul. 14-17). Phytogenic supplementation modulates performance and pathological responses of turkey poult to histomoniasis induced via lateral transmission of *Histomonas meleagridis* in battery cages [Poster Presentation]. Poultry Science Association Annual Meeting, Raleigh, NC, USA.
 10. Ribeiro, G. C., **Niraula, A.**, Calik, A., Blue, CEC., Wagner, A.L., Médina, B., Girard, I., Dalloul, R.A. Phytogenic blend positively influence ileal and cecal microbiota during a necrotic enteritis challenge [Poster Presentation]. Poultry Science Association Annual Meeting, Raleigh, NC, USA.
 11. Dalloul, R., **Niraula, A.**, & Fuller, A. (2025). Applications of a lateral transmission model of *Histomonas meleagridis* to mitigate blackhead disease in turkeys. In Western Poultry Disease Conference (p. 39).
 12. **Niraula, A.**, Taylor Jr, R. L., Fulton, J. E., Wolc, A., Dalloul, R.A. (2025, Jan. 27-28). Chicken alloantigen systems MHC-B, D, E, and I haplotypes are associated with resistance to coccidiosis [Oral Presentation]. International Poultry Scientific Forum, Atlanta, GA, USA.
 13. **Niraula, A.**, Calik A., Perrin, G., Blue, CEC., Dalloul, R.A. (2024, July 15-19). Investigation of intestinal cell autophagy markers in birds infected with *Eimeria maxima*, *Clostridium perfringens*, or both [Poster Presentation]. Poultry Science Association Annual Meeting, Louisville, KY, USA.
 14. **Niraula, A.**, Taylor Jr, R. L., Fulton, J. E., Dalloul, R.A. (2024, Jan. 29-30). Major histocompatibility complex B15 haplotype and alloantigen types D and E exhibit association with resistance traits to coccidiosis challenge in chickens [Oral Presentation]. International Poultry Scientific Forum, Atlanta, GA, USA.
 15. **Niraula, A.**, Fenster, D.A., Wagner, A.L., Médina, B., Girard, I., Fuller, A.L., Dalloul, R.A. (2023, Jul. 10-13). Protective effects of Alterna® HTS in turkey poult raised in a floor pen lateral transmission model of *Histomonas meleagridis* [Oral Presentation]. Poultry Science Association Annual Meeting, Philadelphia, PA, USA.
 16. **Niraula, A.**, Blue, C.E.C., Fenster, D.A., Emami, N.K., Dalloul, R.A. (2023, Jan. 23-24). Assessment of mRNA abundance of key cytokines during histomoniasis in turkey poult in a lateral transmission model [Poster Presentation]. International Poultry Scientific Forum, Atlanta, GA, USA.

AWARDS AND SCHOLARSHIPS

- **Travel Support, 2026**

Awarded a travel scholarship from the Southern Conference of Avian Diseases to present my research findings at the 2026 International Poultry Scientific Forum

- **Travel Support, 2025**

Awarded a travel scholarship from the American Association of Avian Pathologists (AAAP) to present my research findings at the 2025 AAAP Annual Meeting

- **Travel Award, 2025**

Awarded a travel scholarship from the Graduate School at the University of Georgia to present my research findings at the 2025 American Association of Avian Pathologists Annual Meeting

- **Research Excellence Award, 2024**

Awarded first-place prize for a Poster presentation at the Graduate Research Forum organized by the Department of Poultry Science, University of Georgia

- **Climate Action Travel Grant, 2014**

Awarded a travel grant to participate in the Local Conference of Youth (LCOY 13)

- **Merit Scholarship, Government of Nepal, 2014**

Scholarship to study B.V. Sc. & A. H. at the Agriculture and Forestry University

PROFESSIONAL ACTIVITIES

1. **Peer Review**, Poultry Science (Elsevier) – 3 and Frontiers in Immunology – 1
2. **Poster Judge**, Academic and Career Planning (POUL 1075) course at Department of Poultry Science, University of Georgia (2025, October)
3. **Member**: Poultry Science Association, World Poultry Science Association, and American Association of Avian Pathologists
4. **Member**: Poultry Science Graduate Club at the University of Georgia
5. **Event co-ordinator**: Nepalese Student Association at the University of Georgia (2023-2024)
6. **Treasurer**: Climates Nepal (2015-2016)

MENTORING

1. Mentoring two undergraduate students in lab techniques, data documentation, and analysis at the University of Georgia
2. Mentored an undergraduate student and a high school student through the Rising Scholar and Emerging Scholar program, respectively, at the University of Georgia (2024, June-July)